

J1 & J2. Evaluation of Model

Due to our Mentor being absent for an extended period of time, we weren't able to physically work on our project but we continued to create plans and communicate with the group.

Firstly, we created a list of materials that we would need to work on our model. We listed down all the materials using our 3D model and our actual prototype as a guide. We organized the materials based on the availability. We listed them based on whether we had the materials already, if our mentor's were donating / providing us the materials, and whether we had to purchase them. This way, we had a list of materials prepared for our Mentors to review.

We communicated with our Mentors, Ms. R.N and Mr. T.B, about the materials we were planning to use for our project and asked them for recommendations on specific types of materials that they would recommend and also their opinions on these materials.

Since we didn't have any for our project done, our mentors evaluated our scaled version model of our project and also evaluated and read through our plans, material lists. Furthermore, based on that they started to give us feedback for what we will be doing in our final prototype and made suggestions that we can add to our plan/organizer that we had created for the building process of our desk.

- **Mr. R.N (Engineering Teacher/Founder of Construction Club/Many years of experience when it comes to building furniture/working with resin):** He gave ideas on how to design the resin of our build so that it would be aesthetic and some other ideas on what we could change on the structural integrity of the desk itself in order to ensure that it's both durable and stable withstand any situations when the desk is in use by Ms.A.V .
 - He had the concern of how we were going to attach the cabinets with a desk made out of resin. Due to us having a resin design we would have to cut the wood in half to properly have the waterfall/ river effect of resin, but also be able to effectively attach the cabinets for the desk.
 - Another concern for stability wise it would be better to do a waterfall on both sides for the desk to be stable and over balance.

Mr.T.B was our external field expert:

Olivia's Father/ Mr.T.B (Mechanical Engineer/Woodworker/3 decades of experience): He gave advice on how to build our design, especially focused on the structural integrity of the desk and pull-out drawers. We are also about materials we can use that are effective, but also affordable. Finally, he gave us ideas on the tests we can do to test the needs and make sure the prototype meets the requirements.

- A concern of Mr.T.B was our safety. Knowing this is one of the first times we are building a desk and working with a lot of wood, he was concerned for us using sharp tools, saws, and screws. Overall, he had a concern for our health and made sure to guide us a bit in how to use the tools to ensure that no one is harmed during the building process and that the tools are used carefully.

Furthermore both our main mentors and external field experts have guided throughout our design process. After showing the initial 3D model, they gave us some feedback on how we could apply our ideas to an actual model. Mr. R.N had advice on how we could do the Resin River on our desk, suggesting that we create a resin waterfall in order to make it more aesthetically pleasing. Mr. T.B had suggestions on how we could create the cabinets of the desk, giving us advice on the different materials that we could use and advice on how to create it. That was just a few examples of their feedback. They gave us a lot of advice on our 3D model and our prototype model, allowing us to create a list of materials and suggestions that we kept in mind as we continued onto the final model.

Mr. R.N was especially helpful with providing us materials for our model. He provided us with the walnut wood that we were going to use for the desk (\$200 in total) and the drawer slides that we would use for the cabinets.

Mr.T.B is providing us with a lot of tools and power tools to use when building the desk and giving us a proper space that has a lot of room to work and build the desk.

All these suggestions, feedback, and concerns have affected our project and our final prototype of our desk because it has affected the materials we will use, the design itself, structural, integrity, and overall concepts that will be included into the final model of the desk. For example, we initially had a waterfall design for one side of the desk and the other side would be legs. With Mr. R.N's suggestion he insisted that we make both sides of the desk a waterfall for more stability. Furthermore, with these suggestions and feedback going to be implemented into our final model/prototype of our standard computer desk it will result in building a desk that is functional, effective, durable, stabilized, have structural integrity, and overall be a functional desk that pleases Ms.A.V.

Our list of materials is below.

- *Screws / Nails*
- *Dowel Pins*
- *Drawer Rails*
- *Polyethene*
- *Artificial Flower Petals*
- *Table Glides*

Mr. R.N responded to our email with advice on our list of materials and the cost of the wood, which is listed below.

- *Screws / Nails (no nails) wood glue*
- *Dowel Pins (I have them)*
- *Drawer Rails ([suggested above](#))*
- *Polyethene (polyurethane water based fast dry)*
- *Artificial Flower Petals (you choose)*
- *Table Glides (for the bottom?)*
- *Wood would be \$350*

We continued to talk to Mr. R.N on the day we came back from break and we managed to clarify about the price of the materials.

- *For the wood, Mr. R.N would be providing us all the wood for \$200.*
- *Additionally, he would provide us with drawer slides, which would be a price of around \$15-\$25.*

As a group, we bought the resin and flowers that would be used in the resin river. In total, it was \$105.

- *Resin: <https://a.co/d/bV6H0RS>*
- *Flowers: <https://a.co/d/aVXNXUU>*